V4 power and wiring: battery, regulator and schematic

The Mercury V1 sends the servo position signal; a small 2S battery and a 5 V regulator power the SG90. Part sizes below come from the sellers' listings I could reach; I could **not** read the Mercury manual (the Altimeter Cloud site is not reachable from this session), so its servo-header pin order and signal voltage are the items to confirm.

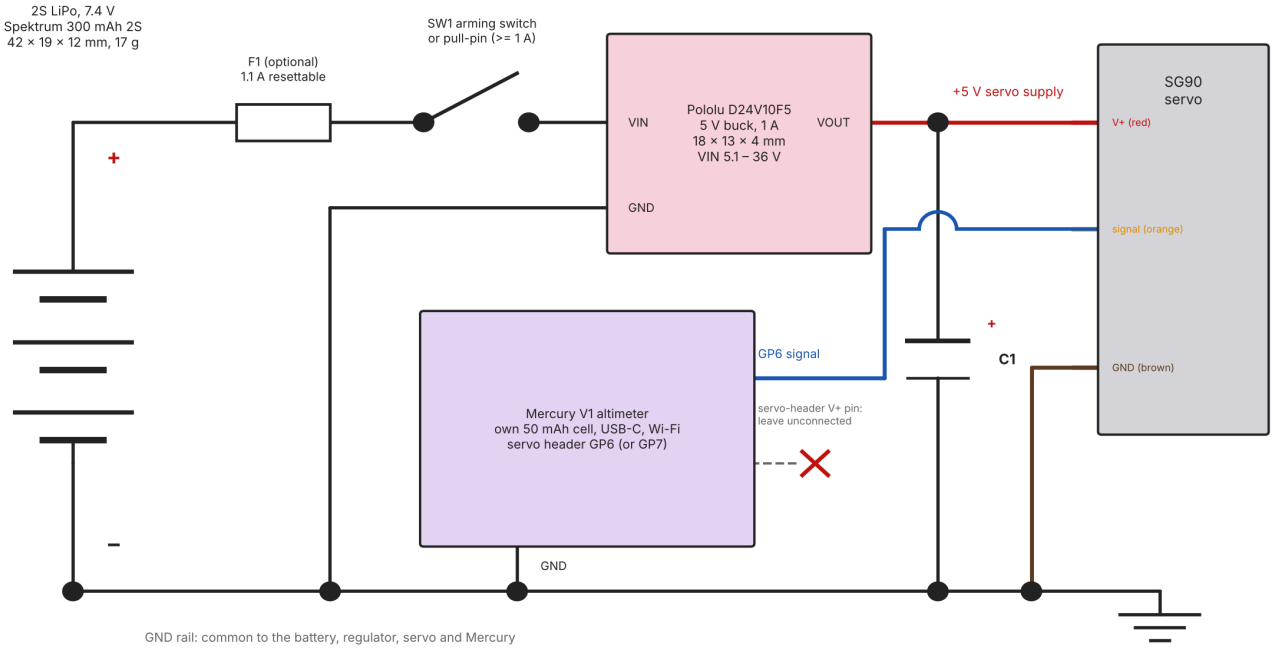
1 Recommended parts

Part	Recommendation	Size / weight (from listings)	Why
Battery	**Spektrum 300 mAh 2S, 30C** (7.4 V nominal, 8.4 V full, PH plug)	42 × 19 × 12 mm, about 17 g (one retailer's listing; measure yours)	Small 2S pack that fits the V4 pocket; 6.0–8.4 V in, which the regulator below accepts; 300 mAh is far more than a flight needs (about 7–12 flights per charge on my estimate)
	Alternatives: Tattu 300 mAh 2S HV; EMAX Tinyhawk S 2S	47 × 17 × 13 mm; 42 × 16.2 × 12.7 mm (listings)	The Tattu is 4.6 mm too long for the new pocket; the EMAX would fit the length and width but is 0.3 mm too tall
Regulator	**Pololu D24V10F5**: 5 V, 1 A step-down	0.5 × 0.7 × 0.14 in (about 13 × 18 × 4 mm), 1 g	Input 5.1–36 V; the SG90 takes 4.8–6 V and draws roughly 0.3–0.55 A at stall (sources vary), so 1 A is about 2× headroom
	Option for more torque: D24V10F6 (6 V, 1 A)	same footprint	SG90 torque rises with voltage (one source: 1.2 → 1.6 kg·cm from 4.8 → 6 V; another lists 1.8 kg·cm at 4.8 V), but this board needs at least 6.1 V in: keep the pack above about 6.5 V
Capacitor C1	470–1000 µF, 10 V, low-ESR electrolytic	about 6.3 mm diameter × 11 mm	Absorbs the servo's current spikes so the regulator and the Mercury do not brown out
Arming	SPST switch ≥ 1 A, or a JST pull-pin / jumper	small slide switch	Keeps the servo dead while you handle it; reach it through the transition ring
Fuse (optional)	1.1 A resettable (polyfuse)	—	Limits a short; the LiPo can deliver far more than the wires should carry
Wire and plugs	24 AWG silicone for power, 28 AWG for the signal; JST-PH 2-pin on the battery (match the plug on the pack you buy)	—	Flexible, heat-tolerant, small

Estimated load: the servo idles near 10 mA and draws about 100–250 mA while moving; one deployment is under a second. A 300 mAh pack therefore lasts many hours of standby (estimate: the regulator's own draw I could not confirm).

2 Wiring schematic

V4 airbrake: servo power and signal (wiring schematic)



C1 = 470 - 1000 uF, 10 V, low-ESR, wired right at the servo pins. The regulator powers the servo; the Mercury only sends the position signal.

Ground is shared. Never join the servo red wire to a Mercury pin; confirm the Mercury servo-header pin order in its manual before wiring.

Servo power and signal. The Mercury has its own 50 mAh battery and does not power the servo.

How it is wired

- **Power path:** battery + → fuse → arming switch → regulator VIN; regulator VOUT (+5 V) → servo red wire and C1; regulator GND, C1 and the servo brown wire all return to the battery –.
- **Signal path:** one Mercury servo header (GP6 or GP7, set in the Mercury configuration as the air-brake servo channel) → servo orange wire. The Mercury GND goes to the same ground rail.
- **The one rule:** the servo's red wire takes power from the regulator only. If the Mercury servo header has a power pin, leave it unconnected (pull the red wire out of the Mercury-side connector shell, or use a servo lead with the red wire cut on that side). Two supplies tied together can damage the Mercury or the regulator.
- **Order of assembly:** bench-test with the servo and a 5 V supply first, then the regulator on the battery, with the switch off until the last step.

3 What I could not verify, and what it means for the CAD

The V4 battery pocket is now sized for the Spektrum 300 mAh 2S. Pocket 42.8 × 19.8 × 12.4 mm (0.4 mm clear per side, 0.2 mm above and below the 42 × 19 × 12 mm pack); the sled's lower block grew from 44.8 × 22.8 to 46.8 × 23.8 mm with its corners trimmed to R25.5 so they clear the opened base ring. The floor between the two layers is 1.8 mm thick. (Later the whole battery layer moved down 2.0 mm for wiring headroom: housing 96.3 mm.) **Battery strap.** The pocket is open at the bottom, so a belly band holds the pack: a nylon tie or narrow hook-and-loop strap (up to 4 mm wide, 1.5 mm thick, about 90 mm long) runs under the pack, up the outside of the lower sled block and across a 1 mm groove in its top. Two channels are modelled, at x = ±17.0…21.4 mm (between the upper block and the end wall); use one or both. The small ribs are notched there, and the transition ring has two notches (6.3 mm × 29 mm, 1.8 mm deep) so the band has room under the sled. I checked a 3.4 mm band in the model: it does not touch the sled, the transition ring or the housing. Still to do on your side: **measure the real pack** (the 42 × 19 × 12 mm figure is one retailer's listing) and allow for the plug and the 3-pin balance lead (fold them beside or on the pack).

- **Mercury servo header:** the Altimeter Cloud flight page I found says the two onboard servo headers are GP6 and GP7, but I could not read the pin order or the signal voltage. Check its manual (the "Mercury V1 schematic" page). The SG90 usually accepts a 3.3 V signal when powered at 5 V, but test it on the bench before flying.
- **SG90 figures vary by source:** stall current 0.3 to 0.55 A, torque 1.2 to 1.8 kg·cm at 4.8 V depending on the version. The V3/V4 torque check used 0.176 N·m (1.8 kg·cm) at 5 V; measure your own servo.
- **Regulator input:** Pololu warns of voltage spikes above its 36 V maximum with long leads at high input voltages; with a 2S pack (8.4 V full) this is not a concern. The D24V10F5 needs at least 5.1 V in, so stop flying a 2S pack that has sagged near 6 V.
- **LiPo safety:** the Spektrum is a standard LiPo (charge to 4.2 V per cell; an HV pack such as the Tattu needs a 4.35 V setting); do not discharge below about 3.3 V per cell; keep the pack away from the ejection-charge heat and from crushing; use a fire-safe bag for charging.
- **Altimeter pressure:** the Mercury reads static pressure, so the bay needs vent holes to the outside (not modelled in the CAD).

Sources

- Altimeter Cloud flight page (GP6/GP7 servo headers, air-brake channel) and the Mercury V1 schematic page (not readable from this session).
- Pololu D24V10F5 5 V, 1 A step-down regulator and D24V10F6 6 V version.
- Spektrum 300 mAh 2S listings: Readymade RC. Other packs considered: Tattu 300 mAh 2S listings: GetFPV, Aloft Hobbies, Gens Ace; Turnigy nano-tech 2S 300 mAh: Botland.
- SG90 figures: SG90 datasheet copy and ProtoSupplies.